

PYTHON PROGRAMMING

Your Magic Report Card Helper! ✨



Made Especially for Grade 3 Coding Wizards!

Welcome, Young Coder!

Have you ever wondered how computers remember your name, your favorite games, or your school marks? Today, we are going to learn a super cool language called **Python**!

We will use our coding magic to help a very tired teacher and build our very own **Digital Report Card Helper**. Let's start our exciting journey! 🚀

PAGE 2: MEET KABIR'S BIG PROBLEM!

Meet **Kabir**! Kabir is a smart boy just like you. Today, he noticed something sad at school.

His teacher, **Ms. Sharma**, is sitting at her desk. She is buried under a mountain of **40 paper report cards**! 

She has to write down every single student's name, their math marks, science marks, and grades by hand. She looks so tired and stressed out!

Kabir looks at his teacher and thinks: *"There must be a better way... Can we use computer magic to help her?"* 

What if a computer could do all the hard work for Ms. Sharma? It would save her time, and she could smile again!

 **Kabir's Idea:** *Let's build a magic digital helper using Python to finish all 40 report cards in the blink of an eye!*

PAGE 3: WHAT IS A VARIABLE? THE MAGIC BOX!

To help Ms. Sharma, we need to learn the first big secret of coding: **Variables**!

Think of a Labeled Box!

Imagine you have a plastic toy box. To know what is inside, you stick a neat paper label on it that says "**Blocks**" or "**Dolls**". Now your toys are safe and easy to find!

In Python, a **Variable** is exactly like that labeled box. It is a little storage space in the computer's brain that holds a piece of information for you.

Look at how we make these magic boxes in Python code:

```
student_name = "Aarav"  
maths_marks = 98
```

In this code:

The Label (Variable Name):

student_name

The Value Stored Inside:

"Aarav"

PAGE 4: YOUR SCHOOL LOCKER HOLDS YOUR STUFF!

Let's find variables around your real school to understand them even better!

The Locker Rule

Think of your **school locker**. It has your name or locker number on the outside. Inside, it holds your heavy backpack, books, and jacket safely.

Variables do the exact same thing inside a computer. They hold names and numbers so they don't get lost!

The Lunchbox Analogy

Think about your lunchbox. If the label says "**Paneer Paratha**" or "**Apple Slices**", that is what you expect to find inside when you open it!

Python works the same way: Each variable is like a labeled box that is super easy for the computer to find and open.

Remember: The **label** is the variable's name, and **what's inside** is the data!

PAGE 5: WHAT IS PRINT()? THE LOUDSPEAKER!

Now that we have stored our data inside our magic boxes, how do we show it to the world? We use a special tool called **print()**!

The Principal's Microphone

Imagine your school Principal standing at the podium during morning assembly. When they speak into the **microphone**, the loud speakers announce the news to the whole school!

In Python, the `print()` function is that exact loudspeaker. It takes whatever is hidden inside your variable boxes and **SHOWS** it on the computer screen loud and clear!

Let's look at a code example:

```
print("Aarav got an A+ Grade!")
```

 **The Magic Commas:** When you put commas inside `print()`, they act like kids joining hands! They connect separate words and numbers together into one big beautiful sentence.

PAGE 6: LET'S BUILD A DIGITAL REPORT CARD!

It is time to make our magic helper code! Let's build it step-by-step together.



Step 1: Store a student's name in a variable!

First, we need a box to hold the student's name. Let's create a variable named `name` and put a bright student named Diya inside it.

```
name = "Diya"
```

The computer now has a memory box labeled **name**, and it is holding the words "**Diya**" safely inside it. Easy, right?



Try It Yourself Mental Game!

Imagine you drag and drop your own name into that box instead of "Diya". What happens?

The variable updates **instantly**! It is just like erasing "Diya" from a sticker label and writing your own name with a marker. The box now holds YOU!

PAGE 7: LET'S ADD MARKS!

A report card isn't complete without showing how hard a student worked. Let's add some score magic!

Step 2: Store the marks in a variable!

Now we create a second box. This box will hold numbers, so we will name the label marks.

```
marks = 95
```

Here, marks is our magic box label, and the number 95 is the treasure that goes inside!

? What if Diya scores a perfect 100?

If Diya studies extra hard and gets a perfect score on her math test, we simply change the code to:

```
marks = 100
```

The computer automatically throws away the old number 95 and saves the fresh number 100! Magic!

PAGE 8: SHOW THE RESULT LOUD & CLEAR!

Now we have our two magic boxes ready: name (holding "Diya") and marks (holding 95). Let's call our loudspeaker tool to announce it!

 **Step 3: Use print() to show the result!**

```
print(name, "scored", marks)
```

Look at how the commas hold hands to create this perfect sentence on the screen:

Output: Diya scored 95

 **Quick Check Time!**

If we change the marks box to hold **90** (marks = 90), what will Python print out on the loudspeaker?

A) Diya scored 95

B) Diya scored 90 ✨ (Yes! The output changes with the variable!)

C) marks scored name

PAGE 9: TRY CHANGING THE VALUES!

The absolute best thing about Python variables is that they can change their minds whenever you want them to!

Let's look at how we can reuse the same loudspeaker line for an entirely new student:

```
name = "Aarav"  
marks = 88  
print(name, "scored", marks)
```

Now, what does Python show on the screen? It prints:

Output: Aarav scored 88



Just Like a Teacher Re-checking a Paper

When Ms. Sharma finds an extra mark on an exam paper, she takes her red pen, crosses out the old score, and writes the new one. Variables do the exact same thing!

Every single time you change the value inside a box, Python completely forgets the old thing and **always remembers the latest thing** you put inside it!

PAGE 10: REAL-WORLD MAGIC!

Guess what? You are now a certified coding helper! The simple code you learned today isn't just a game—it is used by real software engineers every day.



The Online School Portal

When your parents log onto the **Indian School Portal** on a phone or tablet to see your report card, they see blocks showing your subjects:

- **Maths:** 98
- **Science:** 97
- **English:** 91

Behind that neat screen, Python-style variables and loudspeakers are doing the exact same work you just mastered!



Congratulations Wizard!

You helped Kabir find a way to save Ms. Sharma from her 40 paper report cards. By using loops and variables, a computer can generate all 40 report cards in less than a single second! ⚡



You finished the lesson! Keep coding, keep exploring, and keep spreading your magic! ✨